Lecture 14, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

2nd Order Linear Homogeneous IVP

$y'' + p\,y' + q\,y = 0; \qquad y(x_0) = \gamma;\ \ y'(x_0) = \delta$

To solve the IVP the first step is to find (any)

two linearly independent solutions, y_1 and y_2, of the equation in an

interval I that includes x_0.

$$W(y_1, y_2)(x) = \det \begin{pmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{pmatrix} = y_1\,y_2' - y_1'\,y_2(x)$$

y_1 and y_2 are linearly independent $\implies W(x) \neq 0\ \forall x \in I$.

y_1 and y_2 are linearly dependent $\implies W(x) = 0\ \forall x \in I$.

$y = c_1 y_1 + c_2 y_2$

$y' = c_1 y_1' + c_2 y_2'$

Apply the initial conditions:

$y(x_0) = c_1 y_1(x_0) + c_2 y_2(x_0) = \gamma$

$y'(x_0) = c_1 y_1'(x_0) + c_2 y_2'(x_0) = \delta$

Solve for c_1 and c_2 and get the solution to the IVP $y = c_1 y_1 + c_2 y_2$.

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Find I, the interval around x_0 where p and q are continuous.

Let's look at the IVP
$$\begin{cases} y'' + py' + qy = 0 \\ y(x_0) = 1 \\ y'(x_0) = 0 \end{cases}$$

y_1 is the solution.

Look at another IVP
$$\begin{cases} y'' + py' + qy = 0 \\ y(x_0) = 0 \\ y'(x_0) = 1 \end{cases}$$

y_2 is the solution.

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$W(y_1, y_2)(x_0) = y_1(x_0)\, y_2'(x_0) - y_1'(x_0)\, y_2(x_0) = 1 \times 1 - 0 \times 0 = 1 \neq$

$0 \implies y_1$ and y_2 are linearly independent and they satisfy the

equation.

$$\begin{cases} y'' + py' + qy = 0 \\ y(x_0) = \alpha \\ y'(x_0) = \beta \end{cases}$$

Solution $y_3 = c_1 y_1 + c_2 y_2$

Ex. $y'' + y = 0$

$$\begin{array}{l|l} y_1 = cos(x) & y_1 = \mathrm{e}^{ix} \\ y_2 = sin(x) & y_2 = \mathrm{e}^{-ix} \end{array}$$

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Constant Coefficient Equations

$y'' + a\,y' + b\,y = 0$ where a and b are constant.

Ex: $y'' - 3\,y' + 2\,y = 0$

Use solution of the form $y(x) = \mathrm{e}^{rx}$

$y' = re^{rx}$ $\qquad$ $y'' = r^2\mathrm{e}^{rx}$

Substitute in the equation: $r^2\mathrm{e}^{rx} - 3r\mathrm{e}^{rx} + 2\mathrm{e}^{rx} = 0$

$r^2 - 3r + 2 = 0$ $\qquad$ (characteristic equation)

$(r-1)(r-2) = 0 \iff r = 1, r = 2$

Substitute in the equation: $r^2\,\mathrm{e}^{rx} - 3\,r\,\mathrm{e}^{rx} + 2\,\mathrm{e}^{rx} = 0$

and since $\mathrm{e}^{Rx} \neq 0$ we just need to solve the algebraic equation

$r^2 - 3\,r + 2 = 0$ (characteristic equation)

The solutions are $r = 1$ and $r = 2$ so two linearly independent

solutions are

$y_1(x) = \mathrm{e}^{x}$; $y_2(x) = \mathrm{e}^{2x}$

A general solution is then $\boxed{y(x) = c_1\,\mathrm{e}^{x} + c_2\,\mathrm{e}^{2x}}$

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Ex: $y'' - y' = 0$

It's constant coefficient, so use $y(x) = \mathrm{e}^{rx}$ and get the

characteristic equation $r^2 - r = 0$

which has solutions $r = 0$ and $r = 1$.

Two linearly independent solutions are

$y_1(x) = \mathrm{e}^{0*x} = 1; \qquad y_2(x) = \mathrm{e}^{x}$

and a general solution to the equation is $\boxed{y(x) = c_1 + c_2\,\mathrm{e}^{x}}$

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Repeated Roots

Ex: $y'' - 4y' + 4y = 0$

Use $y(x) = e^{rx}$ to get $r^2 - 4r + 4 = (r-2)^2 = 0$ which has the

double root $r = 2$

One solution is $y_1(x) = e^{2x}$

A second linearly independent solution is obtained by multiplying

y_1 by x: $y_2(x) = x e^{2x}$

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To prove this trick we start from $y_1 = \mathrm{e}^{2x}$ and search for a solution

in the form $y(x) = v(x)\,\mathrm{e}^{2x}$

$y' = v'\,\mathrm{e}^{2x} + 2\,v\,\mathrm{e}^{2x}$; $\quad y'' = v''\,\mathrm{e}^{2x} + 4\,v'\,\mathrm{e}^{2x} + 4\,v\,\mathrm{e}^{2x}$

Substituting into the original equation:

$$v''\,\mathrm{e}^{2x} + 4\,v'\,\mathrm{e}^{2x} + 4\,v\,\mathrm{e}^{2x} - 4\left(v'\,\mathrm{e}^{2x} + 2\,v\,\mathrm{e}^{2x}\right) + 4\,v\,\mathrm{e}^{2x} = 0$$

$$\rightarrow \quad v''\,\mathrm{e}^{2x} = 0 \quad \rightarrow \quad v'' = 0$$

Integrate to get $v' = A$. Integrate again: $v(x) = A\,x + B$

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Substituting back into $y(x)$:

$$y(x) = v\mathrm{e}^{2x} = (Ax + B)\,\mathrm{e}^{2x} = Ax\,\mathrm{e}^{2x} + B\,\mathrm{e}^{2x}$$

which is equivalent to

$$y(x) = c_1\,\mathrm{e}^{2x} + c_2\,x\,\mathrm{e}^{2x}$$

Complex Conjugate Roots

Ex: $y'' + y = 0$. Use $y(x) = \mathrm{e}^{rx}$ to get $r^2 + 1 = 0$ which has the complex conjugate solutions $r = \pm \mathrm{i}$. We can write the general solution as $y(x) = c_1 \mathrm{e}^{\mathrm{i}x} + c_2 \mathrm{e}^{-\mathrm{i}x}$

Euler's formula: $\mathrm{e}^{\mathrm{i}x} = \cos(x) + \mathrm{i}\sin(x)$ $\mathrm{e}^{-\mathrm{i}x} = \cos(x) - \mathrm{i}\sin(x)$

$\cos(x) = \frac{\mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x}}{2}$

$\sin(x) = \frac{\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}}{2\,\mathrm{i}}$

Using the first two formulas we can rewrite the general solution as

$$y(x) = c_1\,[\cos(x) + \mathrm{i}\,\sin(x)] + c_2\,[\cos(x) - \mathrm{i}\,\sin(x)] =$$

$$(c_1 + c_2)\cos(x) + \mathrm{i}(c_1 - c_2)\sin(x) = A\cos(x) + B\,\sin(x)$$

This is another general solution.

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